

FES SAIS, Morocco

WMO: 601410

Lat: 33.927N

Lon: 4.978W

Elev: 579

StdP: 94.56

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.9	2.1	-5.1	2.6	8.4	-3.5	3.0	9.1	13.0	11.1	11.1	11.4	2.8	180	0.449

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	16.2	40.0	19.7	38.1	19.7	36.1	19.6	22.0	33.9	21.4	33.1	20.7	32.2	4.8	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.5	14.3	25.6	17.9	13.8	25.2	17.1	13.1	24.6	67.2	33.6	64.5	32.9	62.1	32.3	26.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.3	8.5	7.2	DB	-1.7	42.7	1.7	1.4	-2.9	43.7	-3.9	44.5	-4.9	45.3	-6.1	46.3	(4)
(5)			WB	-2.9	23.7	1.7	1.3	-4.1	24.7	-5.1	25.5	-6.1	26.2	-7.3	27.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.9	10.1	11.0	13.4	15.2	19.1	23.4	26.5	26.9	23.0	19.8	14.3	11.3	(6)	
	DBStd	6.60	2.38	2.56	2.73	2.87	3.83	3.66	3.45	3.25	3.04	3.18	3.26	2.46	(7)	
	HDD10.0	66	27	17	4	1	0	0	0	0	0	0	3	14	(8)	
	HDD18.3	1123	254	206	153	101	37	2	0	0	2	21	127	219	(9)	
	CDD10.0	2940	32	44	111	157	281	401	512	524	391	302	132	53	(10)	
	CDD18.3	954	0	0	1	7	60	153	253	266	143	65	6	0	(11)	
	CDH23.3	12368	1	3	56	178	859	2001	3447	3517	1537	695	71	3	(12)	
(13)	CDH26.7	6633	0	1	8	42	387	1064	2068	2089	717	250	8	0	(13)	
(14)	Wind	WSAvg	3.3	3.4	3.5	3.5	3.3	3.3	3.2	3.2	3.1	3.1	3.4	3.4	(14)	
(15) Precipitation	PrecAvg	532	60	72	75	67	38	20	2	3	12	46	63	67	(15)	
	PrecMax	804	193	194	153	169	128	65	9	18	62	143	170	175	(16)	
	PrecMin	280	0	1	9	4	1	0	0	0	0	0	0	0	(17)	
	PrecStd	138	54	53	43	49	32	18	3	5	14	46	46	46	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.1	23.2	27.7	30.2	36.2	40.8	42.8	42.1	38.2	34.0	27.9	23.2	(19)	
		MCWB	12.2	12.4	16.0	16.7	18.0	19.3	19.7	19.5	18.8	16.8	14.9	13.4	(20)	
	2%	DB	19.8	21.0	24.8	27.8	33.7	37.8	40.1	40.0	35.2	31.7	25.2	20.3	(21)	
		MCWB	11.7	12.5	14.8	16.1	17.8	19.1	19.9	19.7	18.8	16.8	14.2	12.1	(22)	
	5%	DB	17.9	19.2	22.9	25.2	31.0	35.1	38.0	38.0	32.9	29.7	23.2	18.8	(23)	
		MCWB	11.1	12.4	14.1	15.3	17.3	19.0	19.9	20.1	18.7	16.6	14.0	12.0	(24)	
	10%	DB	16.1	17.4	20.8	22.9	28.2	32.8	35.9	35.9	30.9	27.4	21.1	17.0	(25)	
		MCWB	10.8	11.7	13.6	15.1	16.9	18.6	19.8	19.9	18.6	16.5	13.7	11.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.4	15.1	18.0	19.2	20.2	21.9	22.9	23.2	21.6	19.6	18.5	15.6	(27)	
		MCDB	18.4	19.7	24.1	25.9	30.5	33.6	35.0	34.9	31.6	27.6	22.4	19.7	(28)	
	2%	WB	13.1	14.0	16.4	17.6	19.1	20.9	22.0	22.2	20.6	18.7	16.4	14.1	(29)	
		MCDB	17.3	18.6	22.2	24.7	29.2	32.5	34.3	33.9	30.0	26.2	21.9	17.7	(30)	
	5%	WB	12.1	13.1	15.2	16.4	18.2	20.1	21.3	21.5	19.9	18.0	15.2	13.0	(31)	
		MCDB	16.0	17.6	20.8	23.3	28.1	31.4	33.7	33.2	28.9	25.1	20.6	16.8	(32)	
	10%	WB	11.2	12.2	14.1	15.3	17.4	19.3	20.6	20.7	19.3	17.4	14.2	12.1	(33)	
		MCDB	15.2	16.5	19.3	21.4	26.4	30.1	32.8	32.3	28.1	24.4	19.4	15.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	11.3	12.0	12.7	13.0	14.1	15.5	16.5	16.2	14.0	13.0	11.4	11.2	(35)	
		MCDBR	15.1	15.6	17.0	17.9	19.2	18.9	18.4	18.5	17.4	16.7	14.8	14.1	(36)	
	5% WB	MCWBR	9.1	9.3	9.5	8.7	7.5	6.7	6.5	6.6	6.6	7.1	7.5	8.3	(37)	
		MCDWR	11.6	13.3	14.4	15.2	16.5	16.7	16.7	16.3	14.3	13.2	11.7	11.1	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.294	0.329	0.382	0.395	0.428	0.464	0.511	0.486	0.430	0.387	0.333	0.303	(40)	
		taud	2.540	2.434	2.266	2.251	2.179	2.069	1.927	2.005	2.191	2.315	2.472	2.540	(41)	
	Ebn at Noon		927	920	889	892	862	827	786	799	830	840	867	894	(42)	
		Edh at Noon	81	100	129	137	149	166	190	173	137	112	86	77	(43)	
(44) All-Sky Solar Radiation	RadAvg		3.00	3.85	5.03	6.00	6.80	7.70	7.63	6.88	5.63	4.25	3.16	2.77	(44)	
	RadStd		0.33	0.40	0.40	0.43	0.50	0.30	0.19	0.23	0.29	0.31	0.28	0.26	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.51	N/A	N/A	+1.14	+0.37	N/A	N/A	-88	+282	+212
(47)	Regional (6 neighbors)	N/A	N/A	N/A	+1.05	N/A	-0.48	N/A	N/A	N/A	+81

Nomenclature: See separate page